

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Tech	Indian Institute of Technology, Kanpur	8.5/10
2021 - 2025	B.Tech	Gayatri Vidya Parishad College of Engineering	8.96/10
2021	Intermediate	Sri Viswa Junior College	95.2%

Research Experience

- **Ambiguity Minimization in VLMs via Contrastive Classification** — Vision-Language Models (Apr’26–Present)
 - Modeled ambiguous robot instruction following as a **POMDP** and designed a classifier-guided framework for ambiguity-aware decision making.
 - Trained a BCE-based ambiguity classifier on frozen **Qwen2-VL-2B** embeddings and an **AWR** policy for clarification generation and task execution.
 - Created a synthetic dataset pipeline using **GPT-4o** over the **LIBERO** benchmark and evaluated with Exact Match, LLM-as-a-Judge, and decision accuracy.

Key Projects

- **Hybrid Attention Transformer for Semantic Segmentation** — Vision Transformers 🤖 (Mar’26–Apr’26)
 - Designed a **Hybrid Attention Transformer (HAT)** encoder-decoder integrating **Overlapping Cross-Attention**, **Global Sparse Attention**, gated fusion, and channel attention for efficient multi-scale contextual modeling.
 - Built an attention-guided UNet-style decoder with skip-attention gates, boundary refinement, and deep supervision to improve segmentation accuracy and edge localization.
- **Dual-Discriminator GAN with Self-Attention** — PyTorch, Spectral Normalization 🤖 (Sep’25–Nov’25)
 - Designed and implemented a **Dual Discriminator GAN (D2GAN)** with self-attention for CIFAR-10 image synthesis using custom generator and discriminator architectures.
 - Incorporated transposed convolutions, spectral normalization, batch normalization, and a dual-discriminator objective to improve image realism, diversity, convergence, and reduce mode collapse.
- **AI Desktop Assistant with Browser Automation** — AI Agents, FastAPI, Playwright, OpenAI 🤖 (May’26–Present)
 - Developed a modular **agentic desktop assistant** using FastAPI, Playwright, and LLM-based reasoning for browser automation through natural-language instructions.
 - Implemented structured tool-calling and planning pipelines enabling autonomous navigation, content extraction, webpage interaction, and multi-step task execution.
- **Email Spam Detection System** | Python, scikit-learn, TF-IDF, Ensemble Methods 🤖 (May’25–Jul’25)
 - Built a **supervised ML pipeline to classify spam vs. legitimate emails** using a dataset of 6,000 emails with extensive text preprocessing including **tokenization**, **stop-word removal**, **stemming**, and **TF-IDF vectorization**.
 - Benchmarked **Decision Tree and SVM** classifiers and improved performance via **bagging and boosting ensemble methods**, enhancing accuracy and robustness against overfitting.
- **File Compressor and Decompressor** | HTML, CSS, JS 🤖 (Jan’24–Feb’24)
 - Designed and implemented a **file compression and decompression system** to reduce storage size and enable efficient data transmission using **frequency-based encoding** and bit-level operations.

Technical Skills

- **Programming:** C, C++, Python, JavaScript, SQL, HTML.
- **ML / AI:** PyTorch, Vision-Language Models, GANs, Transformers, Semantic Segmentation, C-Learning, scikit-learn, TF-IDF, Ensemble Methods.
- **Software Engineering:** SDLC, OOP, Debugging & Troubleshooting, RESTful API Development, Git/GitHub.
- **Soft Skills:** Leadership, Team Work, Communication, Time Management.

Scholastic Achievements

- Coding Profiles Rating: **Codeforces – 1410, Codechef – 1712, Leetcode – 1783.**
- Secured **All India Rank 403** in **GATE 2025** among 1.75 Lakh candidates.
- Selected for **International Collegiate Programming Contest (Amritapuri Region)** Regionals, 2024.
- Selected for **International Collegiate Programming Contest (Amritapuri Region)** Regionals, 2023.
- Selected for **International Collegiate Programming Contest (Kanpur Region)** Regionals, 2023.

Positions of Responsibility

- **Event Management Lead, Google Developer Student Club** (2023-24)
 - Conducted 2 Speaker Sessions, one coding contest and been a part of core team for Devfest.
 - **Problem setter** for a coding contest.